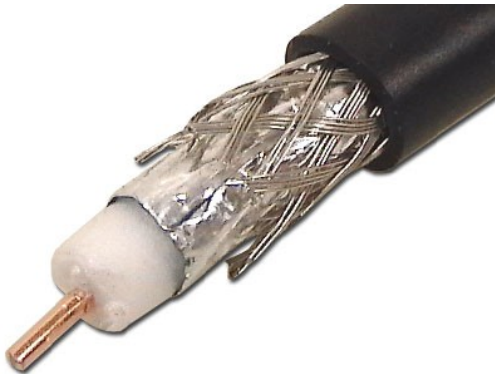


ECE 3317

Dr. Jiefu Chen

Notes 6

Transmission Lines (Time Domain)

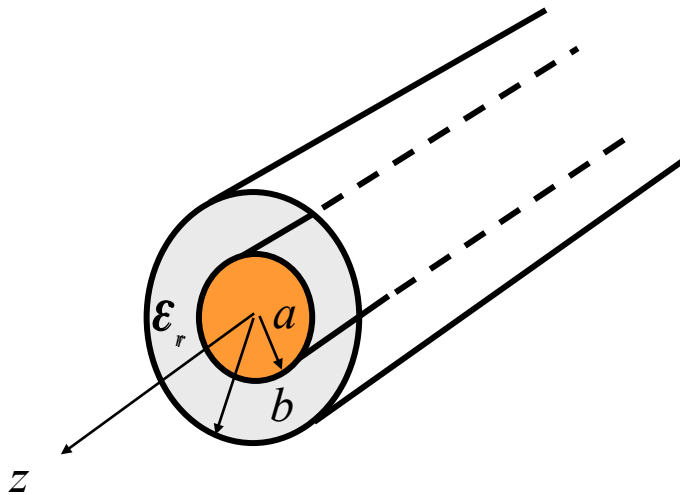


Adapted from notes by Prof. Stuart A. Long

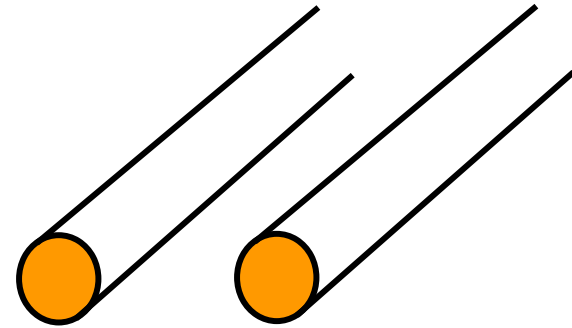
Transmission Lines

A transmission line is a **two-conductor system** that is used to transmit a signal from one point to another point.

Two common examples:



Coaxial cable

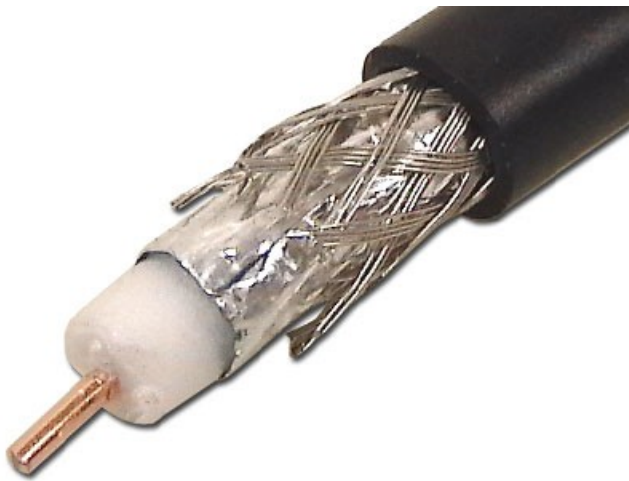


Twin lead

A transmission line is normally used in the **balanced mode**, meaning equal and opposite currents (and charges) on the two conductors.

Transmission Lines (cont.)

Here's what they look like in real-life.

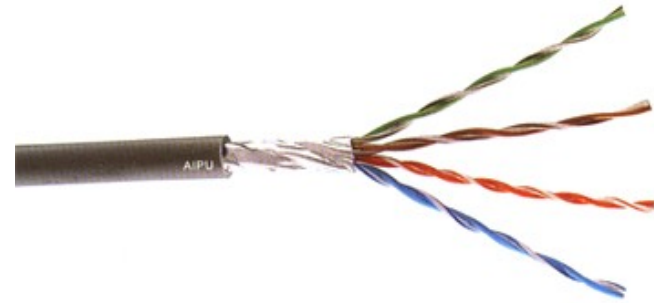


Coaxial cable



Twin lead

Transmission Lines (cont.)



CAT 5 cable
(twisted pair)

Overhead Power Line

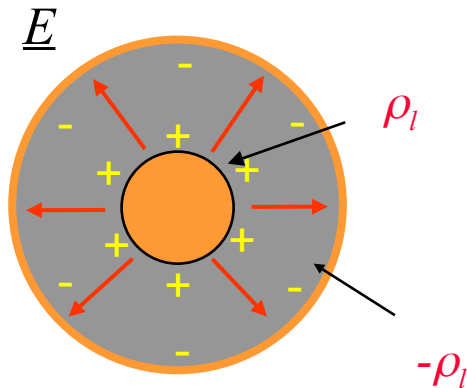
idss.mit.edu



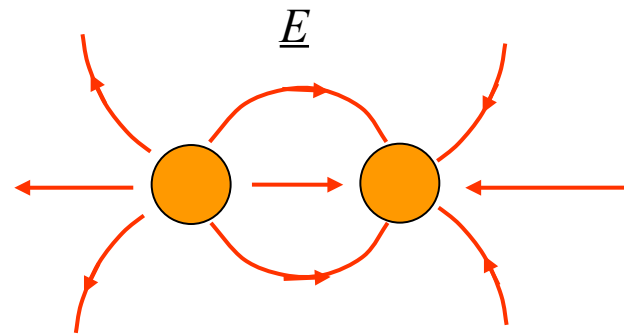
Transmission Lines (cont.)

Some practical notes:

- Coaxial cable is a perfectly shielded system (no interference).
- Twin line is not a shielded system – more susceptible to noise and interference.
- Twin line may be improved by using a form known as “twisted pair” (e.g., CAT 5 cable). This results in less interference.



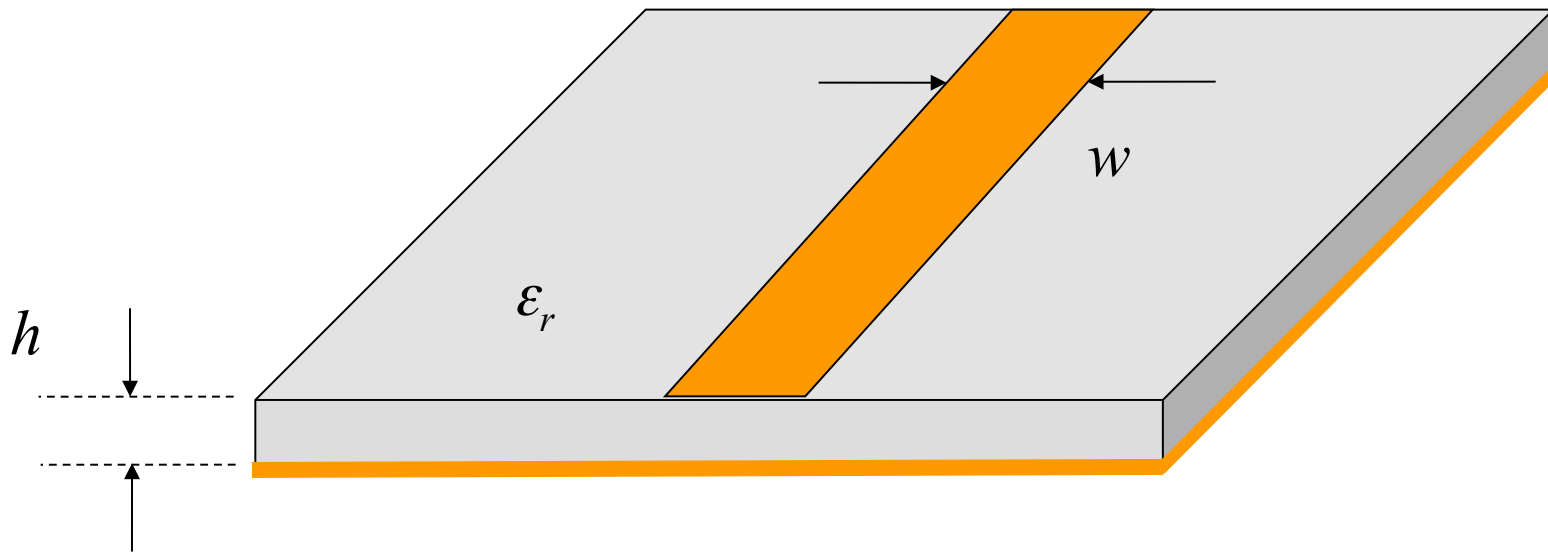
Coax



Twin lead

Transmission Lines (cont.)

Another common example (for printed circuit boards):

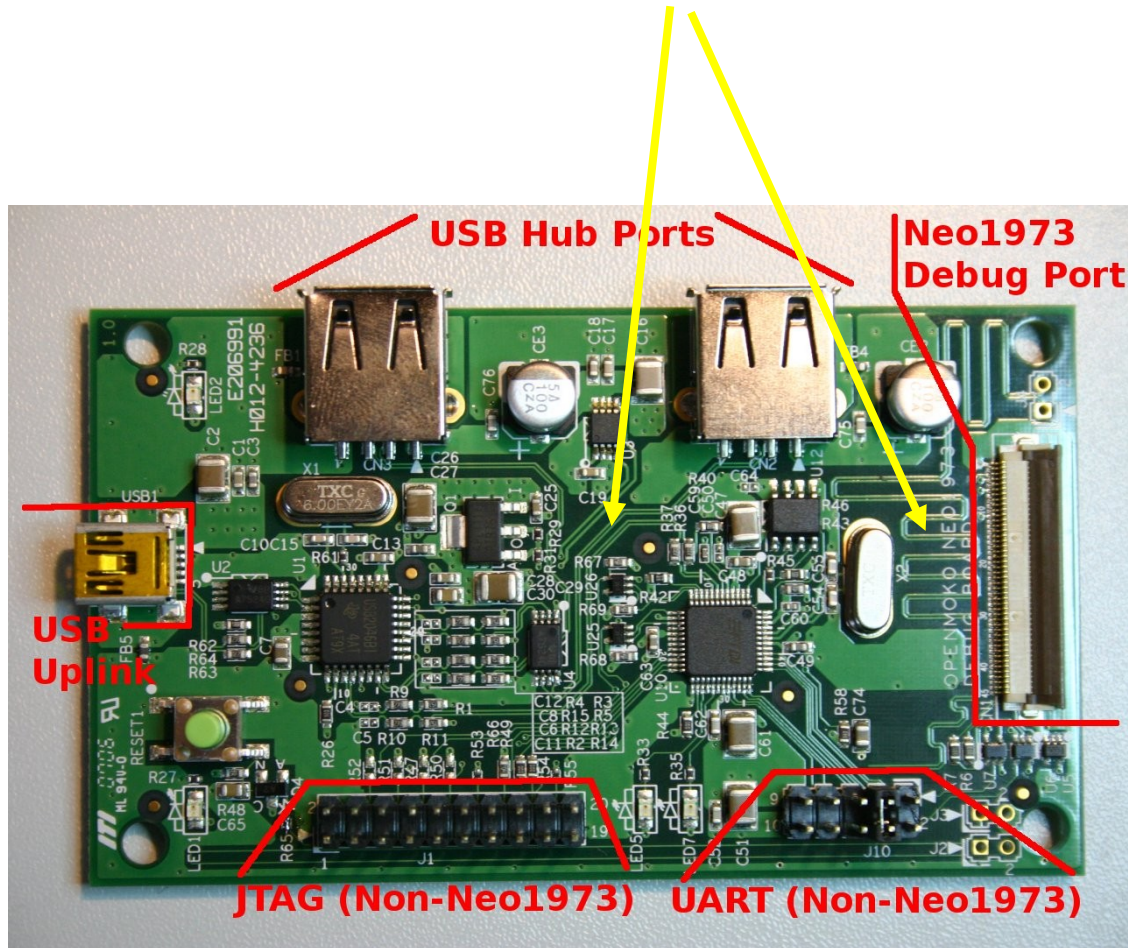


Microstrip line

Transmission Lines (cont.)

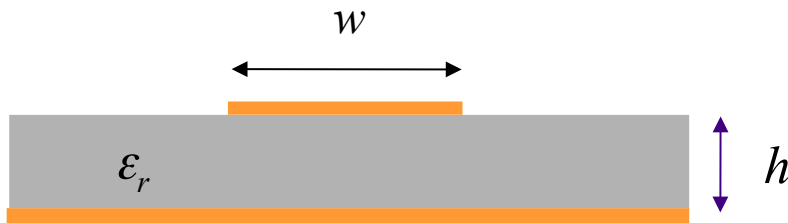
Transmission lines are commonly met on printed-circuit boards.

Microstrip line

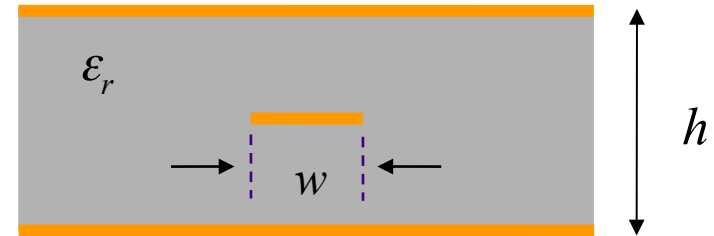


Transmission Lines (cont.)

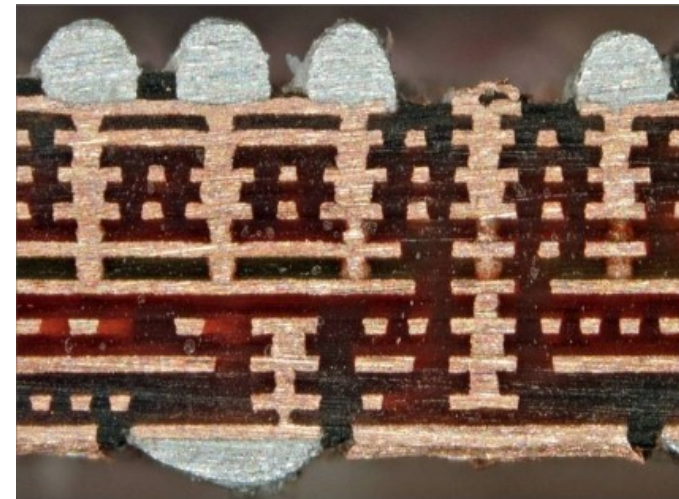
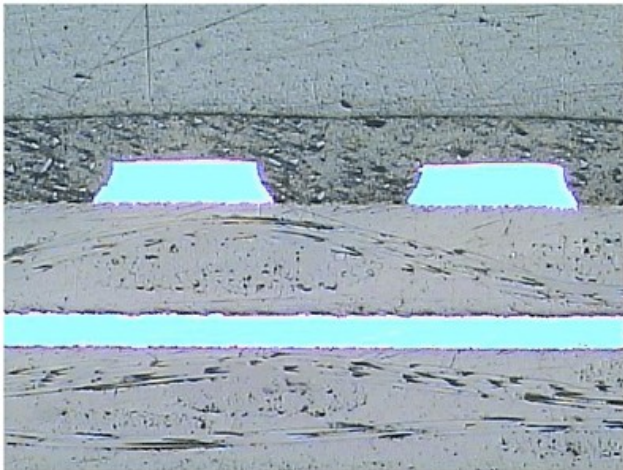
Transmission lines commonly met on printed-circuit boards



Microstrip

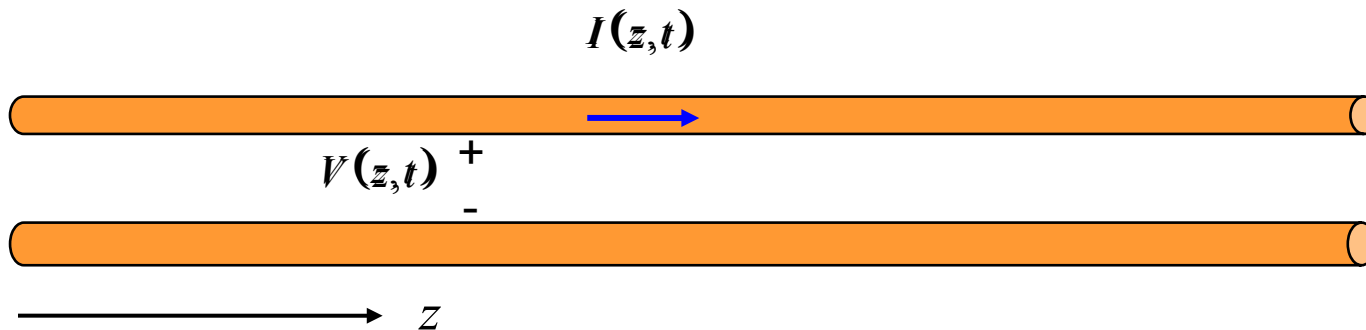


Stripline



Transmission Lines (cont.)

Symbol for transmission line:



Note: We use this schematic to represent a general transmission line, no matter what the actual shape of the conductors.

Transmission Lines (cont.)

Four fundamental parameters that characterize any transmission line:



These are “per unit length” parameters.

C = capacitance/length [F/m]

L = inductance/length [H/m]

R = resistance/length [Ω /m]

G = conductance/length [S/m]

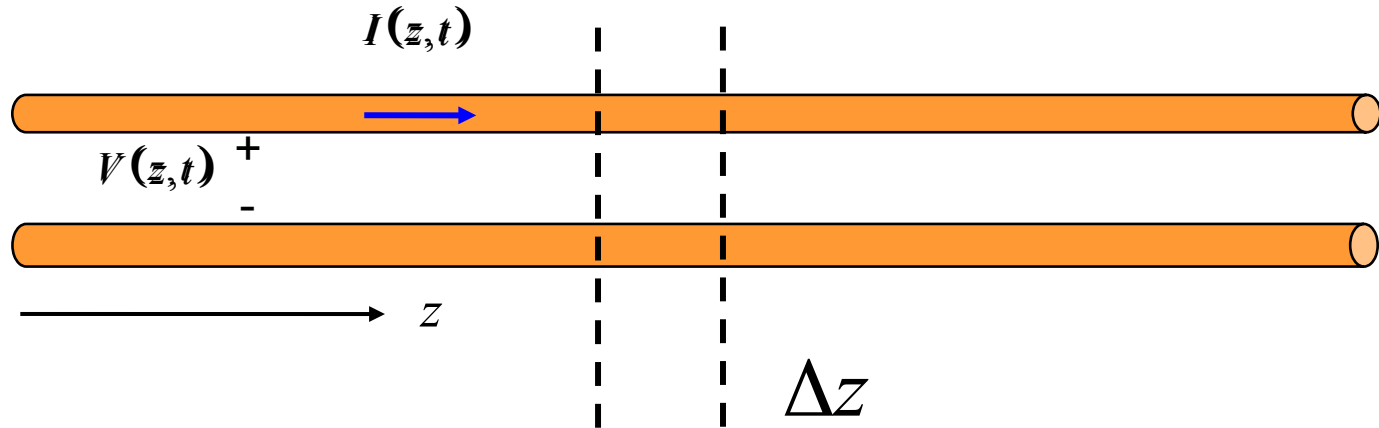
Capacitance between the two wires

Inductance due to stored magnetic energy

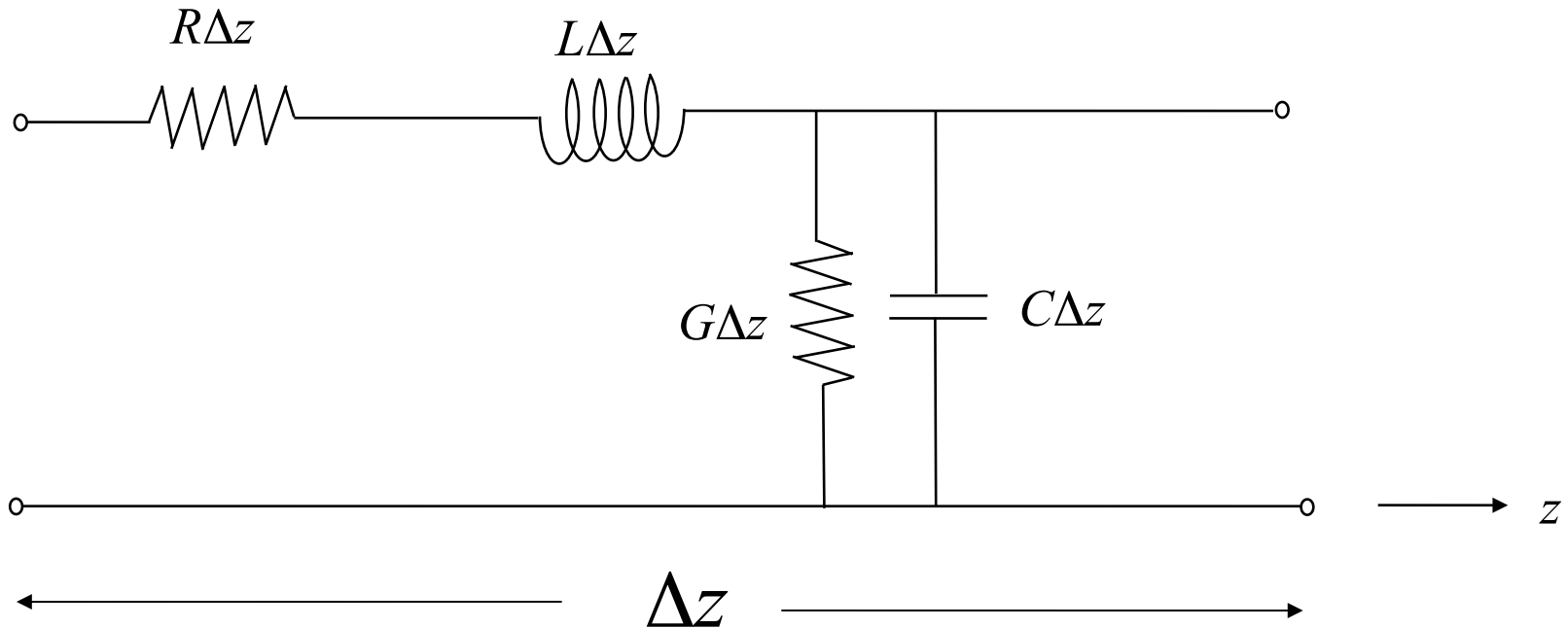
Resistance due to the conductors

Conductance due to the filling material between the wires

Circuit Model

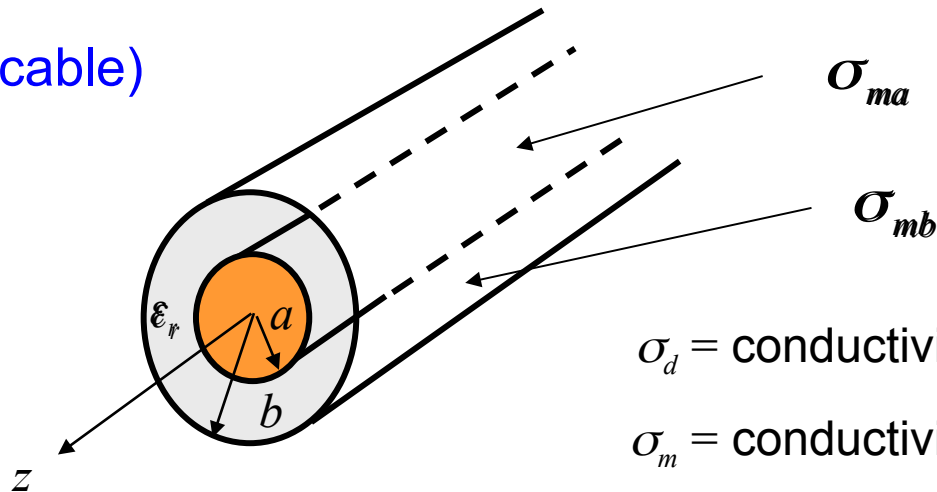


Circuit Model:



Coaxial Cable

Example (coaxial cable)



σ_d = conductivity of dielectric [S/m]

σ_m = conductivity of metal [S/m]

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

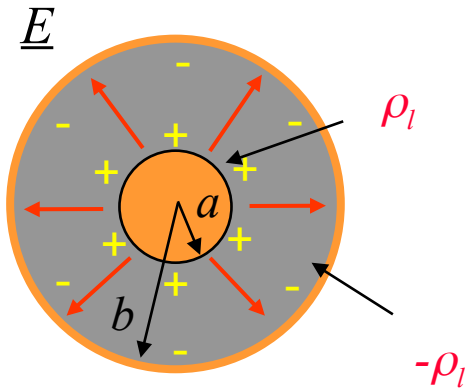
$$R = \left(\frac{1}{2\pi a \delta_a \sigma_{ma}} + \frac{1}{2\pi b \delta_b \sigma_{mb}} \right) \quad [\Omega/\text{m}]$$

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma_m}} \quad (\text{skin depth of metal})$$

$$\delta_{a,b} = \sqrt{\frac{2}{\omega\mu\sigma_{ma,mb}}}$$

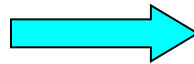
Coaxial Cable (cont.)

Overview of derivation: capacitance per unit length



$$C = \frac{Q}{V} \frac{1}{\Delta z} = \frac{\rho_l}{V}$$

$$V = \int_a^b E_\rho d\rho = \int_a^b \left(\frac{\rho_l}{2\pi\epsilon_0\epsilon_r\rho} \right) d\rho = \frac{\rho_l}{2\pi\epsilon_0\epsilon_r} \ln\left(\frac{b}{a}\right)$$

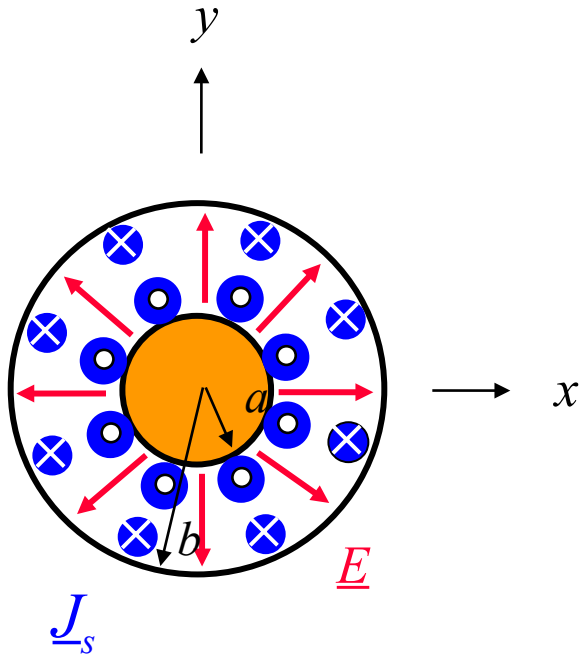


$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)}$$

(suggested reading, not required)

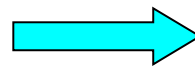
Coaxial Cable (cont.)

Overview of derivation: inductance per unit length



$$L = \frac{\Psi}{I} \frac{1}{\Delta z}$$

$$\begin{aligned}\Psi &= \Delta z \int_a^b \mu_0 H_\phi d\rho = \Delta z \int_a^b \mu_0 \left(\frac{I}{2\pi\rho} \right) d\rho \\ &= \Delta z \mu_0 \left(\frac{I}{2\pi} \right) \ln \left(\frac{b}{a} \right)\end{aligned}$$

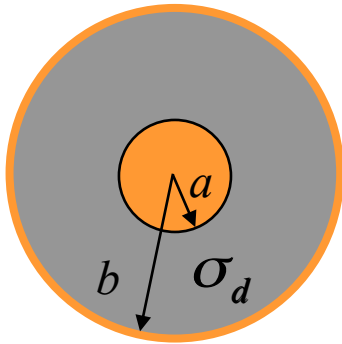


$$L = \frac{\mu_0}{2\pi} \ln \left(\frac{b}{a} \right)$$

(suggested reading, not required)

Coaxial Cable (cont.)

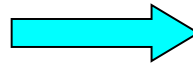
Overview of derivation: conductance per unit length



RC Analogy:

$$\begin{aligned}\epsilon &\rightarrow \sigma_d \\ C &\rightarrow G\end{aligned}$$

$$C = \frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)}$$



$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)}$$

(suggested reading, not required)

Coaxial Cable (cont.)

Relation Between L and C :

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}] \qquad L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$LC = \mu_0\mu_r\epsilon_0\epsilon_r = \mu\epsilon$$

Speed of light in dielectric medium: $c_d = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_r\epsilon_r}} \cdot \frac{1}{\sqrt{\mu_0\epsilon_0}}$

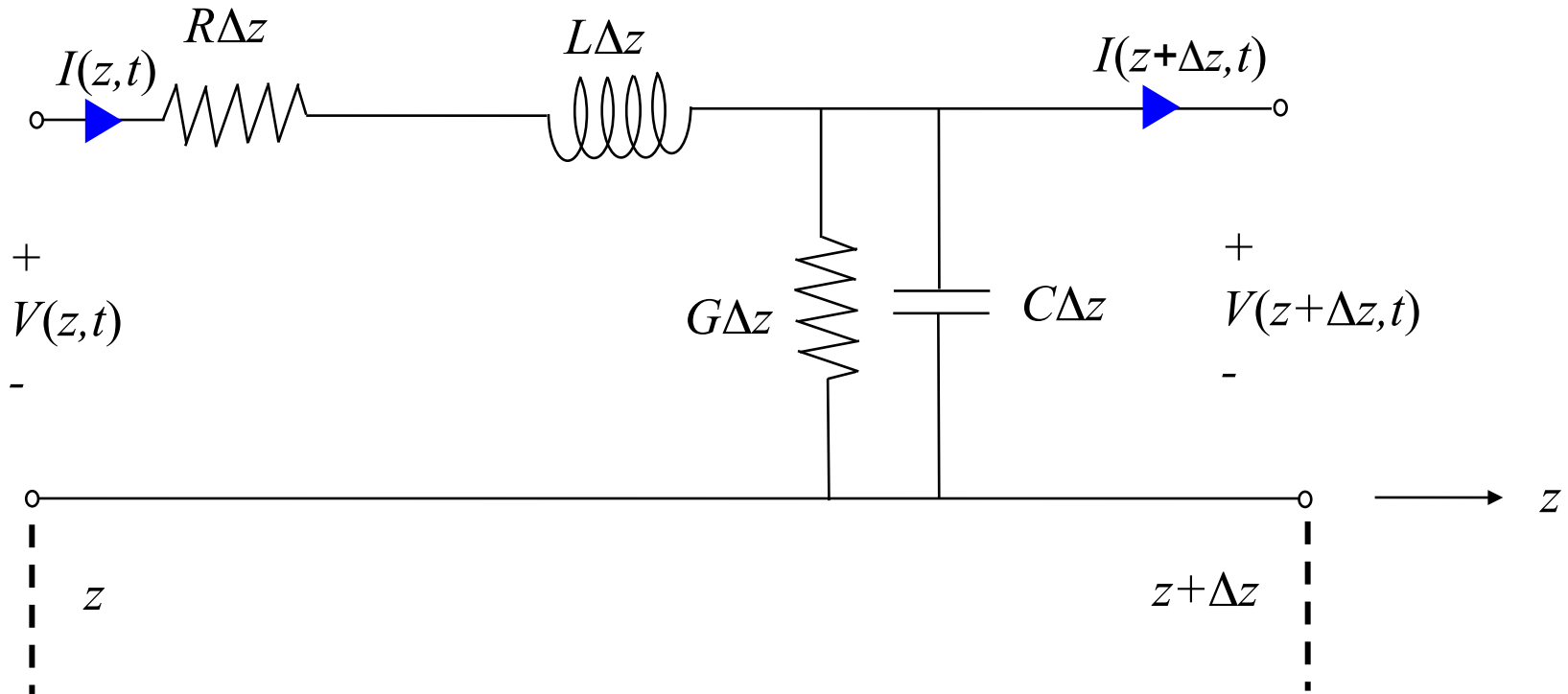
Hence:

$$LC = \frac{1}{c_d^2}$$

This is true for ALL transmission lines.

Telegrapher's Equations

Apply KVL and KCL laws to a small slice of line:



$$\text{KVL: } V(z,t) = V(z + \Delta z,t) + I(z,t)R\Delta z + L\Delta z \frac{\partial I(z,t)}{\partial t}$$

$$\text{KCL: } I(z,t) = I(z + \Delta z,t) + V(z + \Delta z,t)G\Delta z + C\Delta z \frac{\partial V(z + \Delta z,t)}{\partial t}$$

Telegrapher's Equations (cont.)

Hence

$$\frac{V(z + \Delta z, t) - V(z, t)}{\Delta z} = -RI(z, t) - L \frac{\partial I(z, t)}{\partial t}$$

$$\frac{I(z + \Delta z, t) - I(z, t)}{\Delta z} = -GV(z + \Delta z, t) - C \frac{\partial V(z + \Delta z, t)}{\partial t}$$

Now let $\Delta z \rightarrow 0$:

“Telegrapher's Equations (TEs)”

$$\begin{aligned}\frac{\partial V}{\partial z} &= -RI - L \frac{\partial I}{\partial t} \\ \frac{\partial I}{\partial z} &= -GV - C \frac{\partial V}{\partial t}\end{aligned}$$

Telegrapher's Equations (cont.)

- Take the derivative of the first TE with respect to z .
- Substitute in from the second TE.

$$\begin{aligned}
 \frac{\partial^2 V}{\partial z^2} &= -R \frac{\partial I}{\partial z} - L \frac{\partial}{\partial z} \left(\frac{\partial I}{\partial t} \right) \\
 &= -R \frac{\partial I}{\partial z} - L \frac{\partial}{\partial t} \left(\frac{\partial I}{\partial z} \right) \\
 &= -R \left[-GV - C \frac{\partial V}{\partial t} \right] - L \left[-G \frac{\partial V}{\partial t} - C \frac{\partial^2 V}{\partial t^2} \right]
 \end{aligned}$$

$\frac{\partial I}{\partial z} = -GV - C \frac{\partial V}{\partial t}$

Telegrapher's Equations (cont.)

$$\frac{\partial^2 V}{\partial z^2} = -R \left[-GV - C \frac{\partial V}{\partial t} \right] - L \left[-G \frac{\partial V}{\partial t} - C \frac{\partial^2 V}{\partial t^2} \right]$$

Hence, we have:

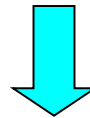
$$\frac{\partial^2 V}{\partial z^2} - (RG)V - (RC + LG) \frac{\partial V}{\partial t} - LC \left(\frac{\partial^2 V}{\partial t^2} \right) = 0$$

There is no exact solution to this differential equation, except for the **lossless case**.

Telegrapher's Equations (cont.)

Lossless case: $R = G = 0$

$$\frac{\partial^2 V}{\partial z^2} - (RG)V - (RC + LG)\frac{\partial V}{\partial t} - LC\left(\frac{\partial^2 V}{\partial t^2}\right) = 0$$



$$\frac{\partial^2 V}{\partial z^2} - LC\left(\frac{\partial^2 V}{\partial t^2}\right) = 0$$

“wave equation”

Note: The current satisfies the same differential equation.

Solution to Telegrapher's Equations

Hence we have

$$\frac{\partial^2 V}{\partial z^2} - \frac{1}{c_d^2} \left(\frac{\partial^2 V}{\partial t^2} \right) = 0$$

Solution:

$$V(z, t) = f(z - c_d t) + g(z + c_d t)$$

where f and g are arbitrary functions.

This is called the D'Alembert solution to the wave equation (the solution is in the form of traveling waves).

Traveling Waves

Proof of solution

$$\frac{\partial^2 V}{\partial z^2} = \frac{1}{c_d^2} \left(\frac{\partial^2 V}{\partial t^2} \right)$$

General solution: $V(z, t) = f(z - c_d t) + g(z + c_d t)$

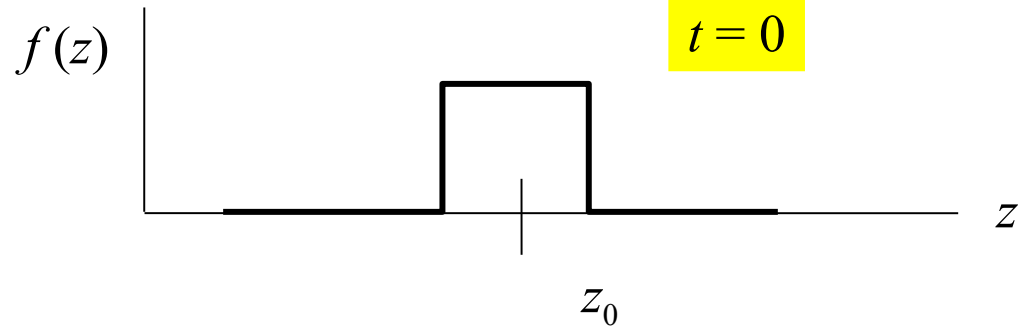
$$\frac{\partial^2 V(z, t)}{\partial z^2} = f''(z - c_d t) + g''(z + c_d t)$$

$$\frac{\partial^2 V(z, t)}{\partial t^2} = (-c_d)^2 f''(z - c_d t) + c_d^2 g''(z + c_d t)$$

It is seen that the differential equation is satisfied by the general solution.

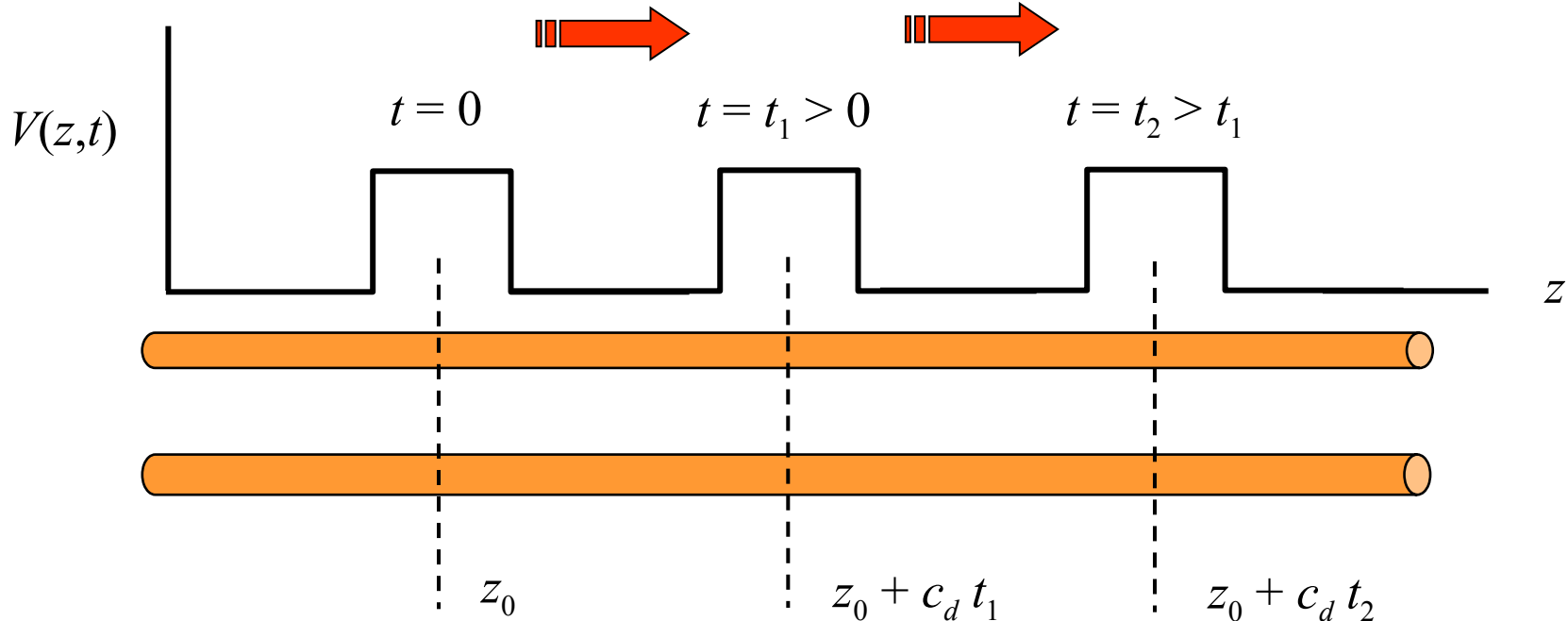
Traveling Waves (cont.)

Example
(square pulse):



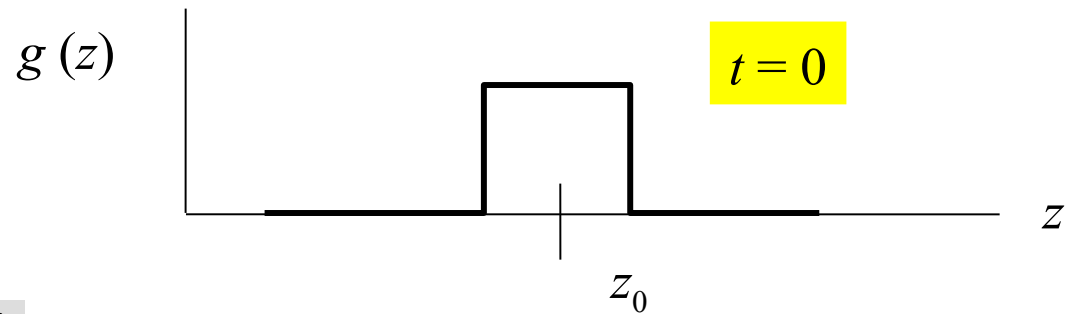
$$V(z, t) = f(z - c_d t)$$

“snapshots of the wave”



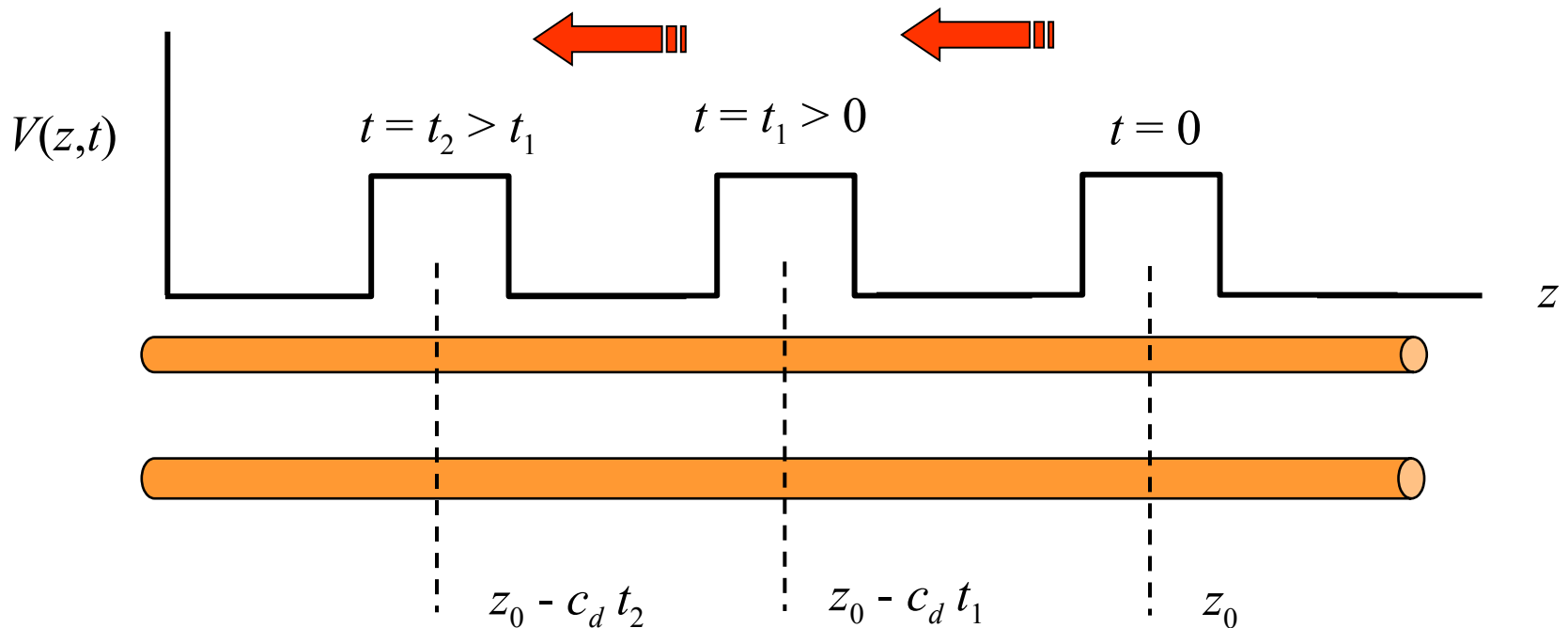
Traveling Waves (cont.)

Example
(square pulse):



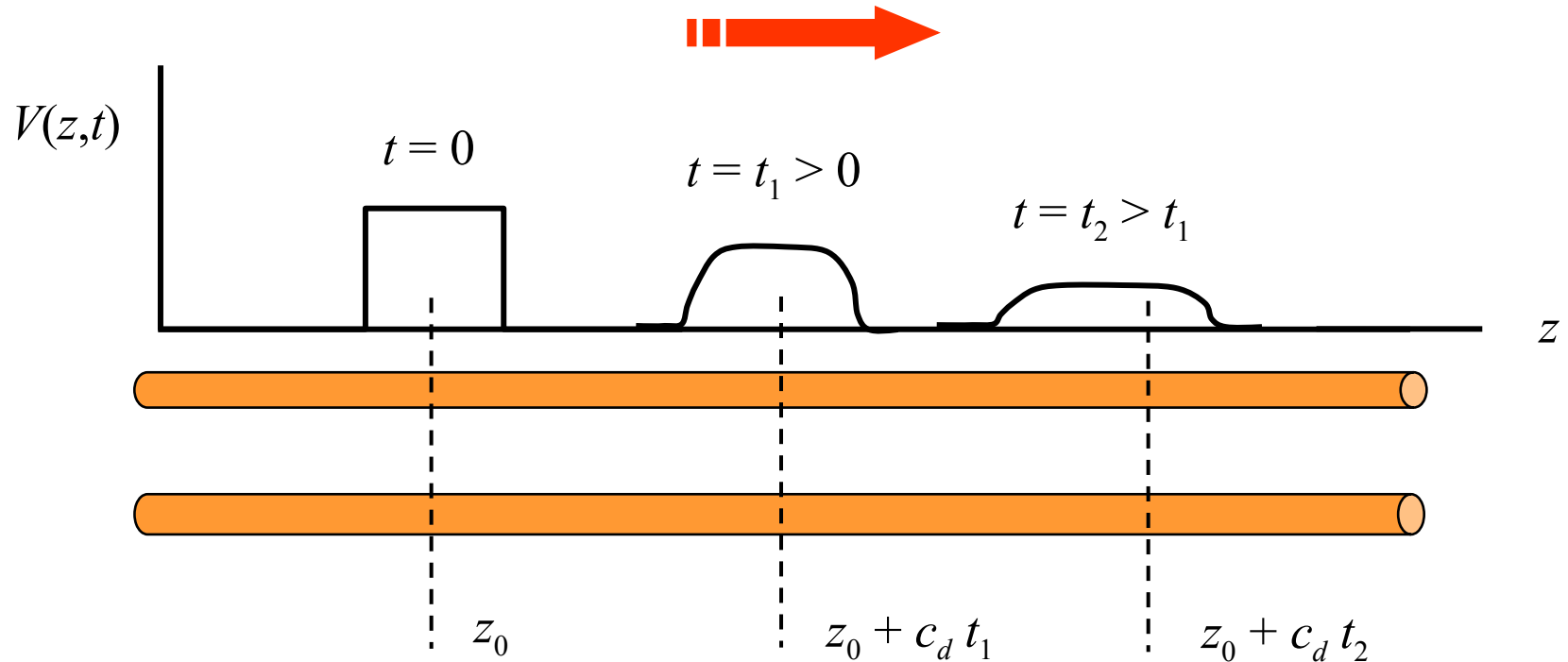
$$V(z, t) = g(z + c_d t)$$

“snapshots of the wave”



Traveling Waves (cont.)

Loss causes an attenuation in the signal level, and it also causes distortion (the pulse changes shape and usually gets broader).



(These effects can be studied numerically, but not here in 3317.)

Current

Our goal is to now solve for the current on the line.

(first Telegrapher's equation)

Lossless

$$\frac{\partial V}{\partial z} = -RI - L \frac{\partial I}{\partial t} \quad \Rightarrow \quad \frac{\partial V}{\partial z} = -L \frac{\partial I}{\partial t}$$

Assume the following forms:

$$V(z, t) = f(z - c_d t) + g(z + c_d t)$$

$$I(z, t) = u(z - c_d t) + v(z + c_d t)$$

The derivatives are:

$$\frac{\partial V(z, t)}{\partial z} = f'(z - c_d t) + g'(z + c_d t)$$

$$\frac{\partial I(z, t)}{\partial t} = -c_d u'(z - c_d t) + c_d v'(z + c_d t)$$

Current (cont.)

$$\frac{\partial V}{\partial z} = -L \frac{\partial I}{\partial t}$$

This becomes

$$f'(z - c_d t) + g'(z + c_d t) = -L [-c_d u'(z - c_d t) + c_d v'(z + c_d t)]$$

Equating like terms, we have

$$f'(z - c_d t) = -L [-c_d u'(z - c_d t)]$$

$$g'(z + c_d t) = -L [c_d v'(z + c_d t)]$$

Hence we have

$$u(z - c_d t) = \frac{1}{L c_d} f(z - c_d t)$$

$$v(z + c_d t) = -\frac{1}{L c_d} g(z + c_d t)$$

Note: There may be a constant of integration, but this would correspond to a DC current, which is ignored here.

Current (cont.)

Observation about term:

$$Lc_d = L \left(\frac{1}{\sqrt{\mu\epsilon}} \right) = L \left(\frac{1}{\sqrt{LC}} \right) = \sqrt{\frac{L}{C}}$$

Define the characteristic impedance Z_0 of the line:

$$Z_0 = \sqrt{\frac{L}{C}}$$

The units of Z_0 are Ohms.

Then

$$u(z - c_d t) = \frac{1}{Z_0} f(z - c_d t)$$

$$v(z + c_d t) = -\frac{1}{Z_0} g(z + c_d t)$$

Current (cont.)

General solution:

$$V(z,t) = f(z - c_d t) + g(z + c_d t)$$

$$I(z,t) = \frac{1}{Z_0} [f(z - c_d t) - g(z + c_d t)]$$

For a forward wave, the current waveform is the same as the voltage, but reduced in amplitude by a factor of Z_0 .

For a backward traveling wave, there is a minus sign as well.

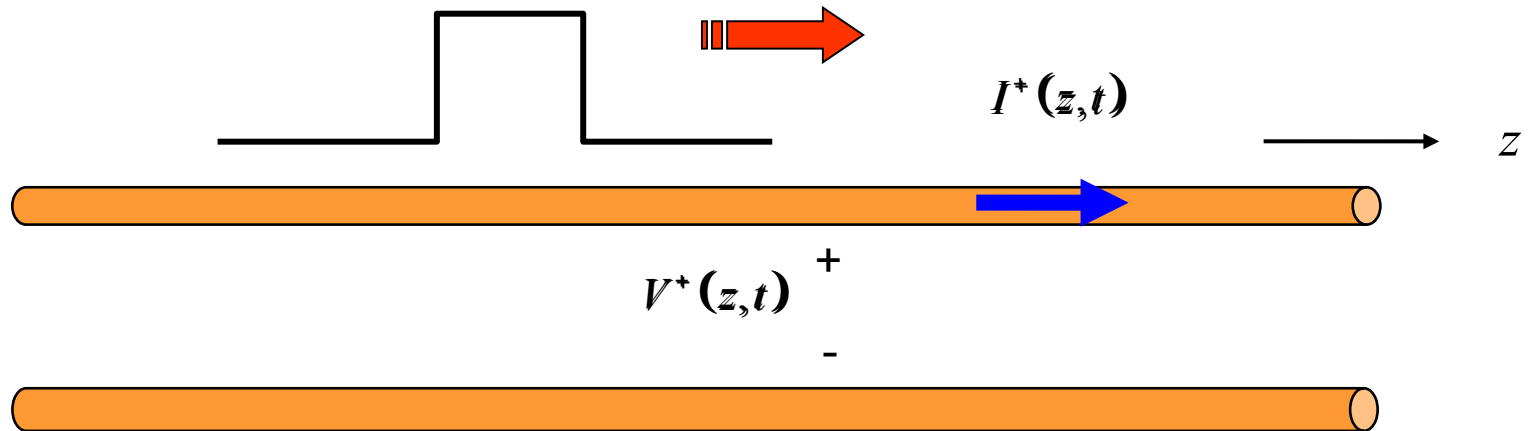
Current (cont.)

Picture for a forward-traveling wave:

$$V^+(z,t) = f(z - c_d t)$$

$$I^+(z,t) = \frac{1}{Z_0} f(z - c_d t)$$

forward-traveling wave



$$\frac{V^+(z,t)}{I^+(z,t)} = Z_0$$

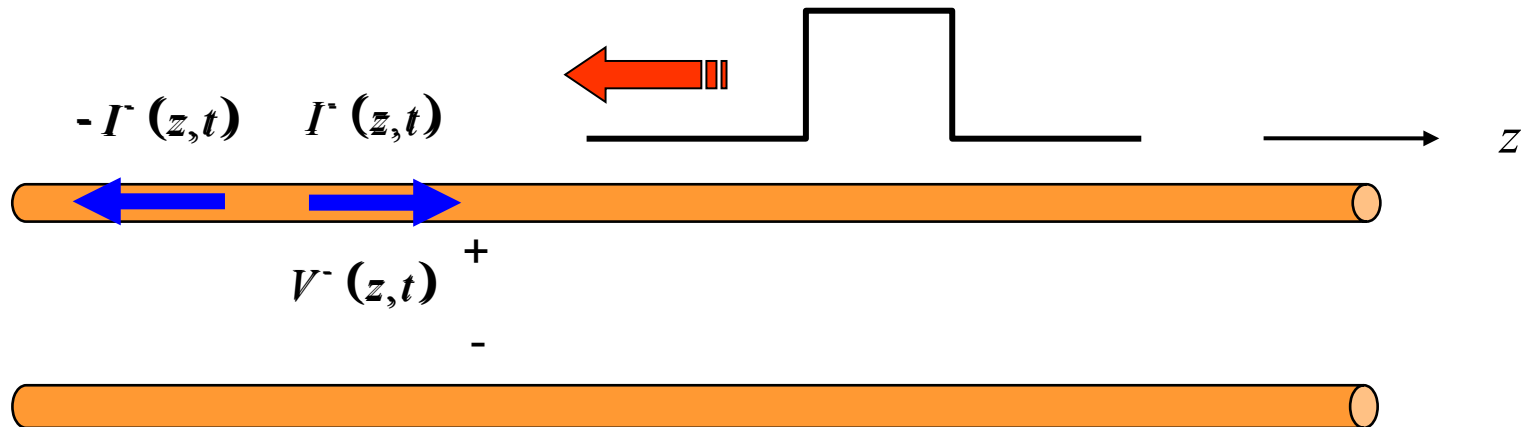
Current (cont.)

Physical interpretation of minus sign for the backward-traveling wave:

$$V^-(z,t) = g(z + c_d t)$$

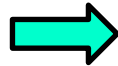
$$I^-(z,t) = -\frac{1}{Z_0} g(z + c_d t)$$

backward-traveling wave



The minus sign arises from the reference direction for the current.

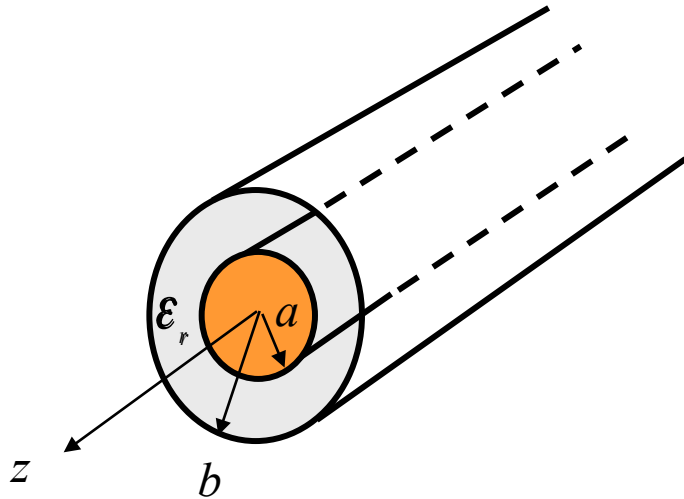
$$\frac{V^-(z,t)}{-I^-(z,t)} = Z_0$$



$$\frac{V^-(z,t)}{I^-(z,t)} = -Z_0$$

Coaxial Cable

Example: Find the characteristic impedance of a coax.



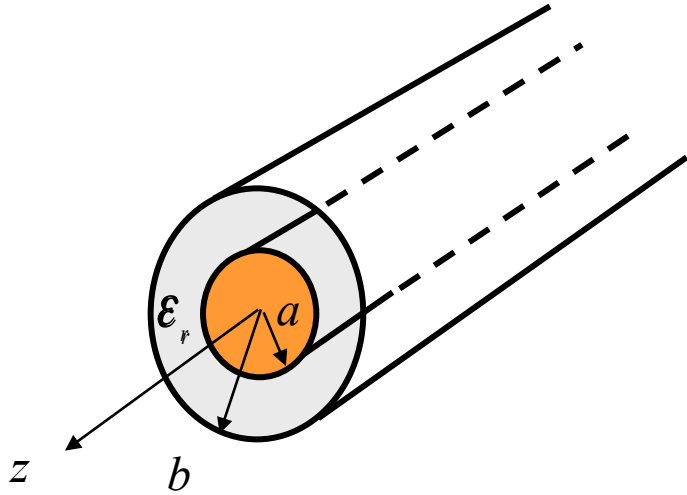
$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{\frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)}{\frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)}}}$$

$$Z_0 = \frac{1}{2\pi} \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{1}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

Coaxial Cable (cont.)



$$Z_0 = \frac{1}{2\pi} \eta_0 \frac{1}{\sqrt{\epsilon_r}} \ln \left(\frac{b}{a} \right) \quad [\Omega]$$

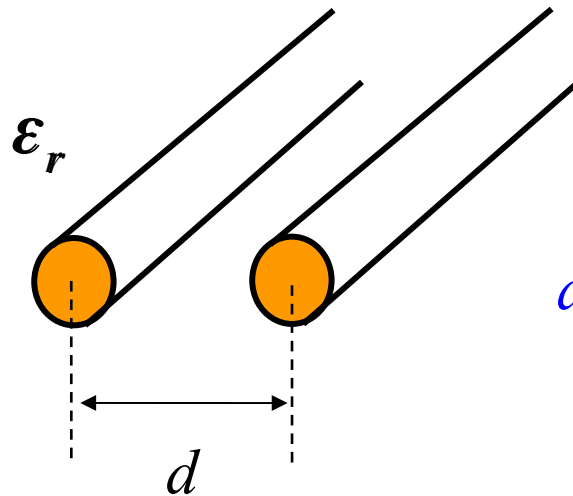
$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \quad (\text{intrinsic impedance of free space})$$

$$\epsilon_0 \doteq 8.8541878 \times 10^{-12} \text{ [F/m]}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]} \text{ (exact)}$$

$$\eta_0 \doteq 376.7303 \text{ } [\Omega]$$

Twin Lead



a = radius of wires

$$C = \frac{\pi \epsilon_0 \epsilon_r}{\cosh^{-1} \left(\frac{d}{2a} \right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{\pi} \cosh^{-1} \left(\frac{d}{2a} \right) \quad [\text{H/m}]$$

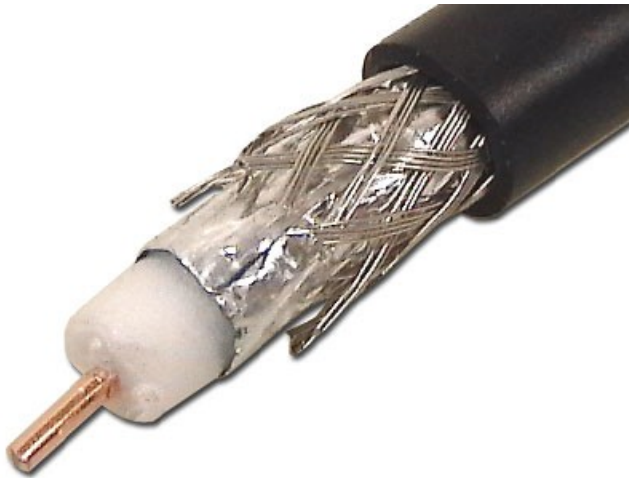
$$Z_0 = \frac{1}{\pi} \eta_0 \frac{1}{\sqrt{\epsilon_r}} \cosh^{-1} \left(\frac{d}{2a} \right) \quad [\Omega]$$

$$Z_0 \approx \frac{1}{\pi} \eta_0 \frac{1}{\sqrt{\epsilon_r}} \ln \left(\frac{d}{a} \right) \quad [\Omega]$$

$a \ll d$

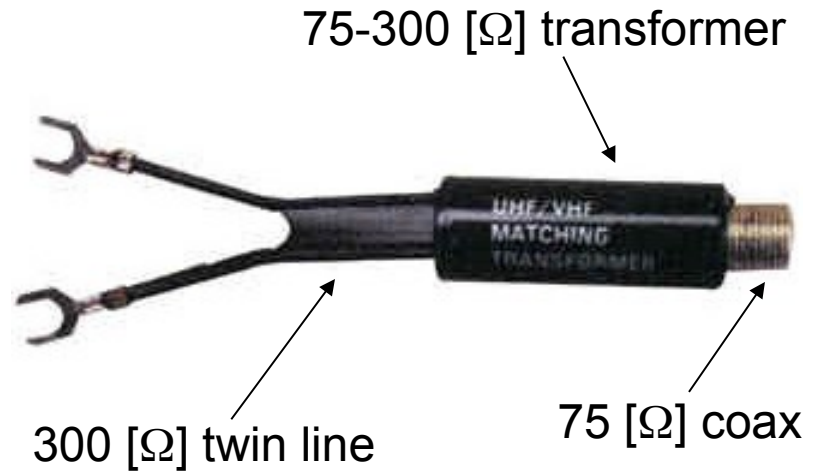
Twin Line (cont.)

These are the common values used for TV.



Coaxial cable

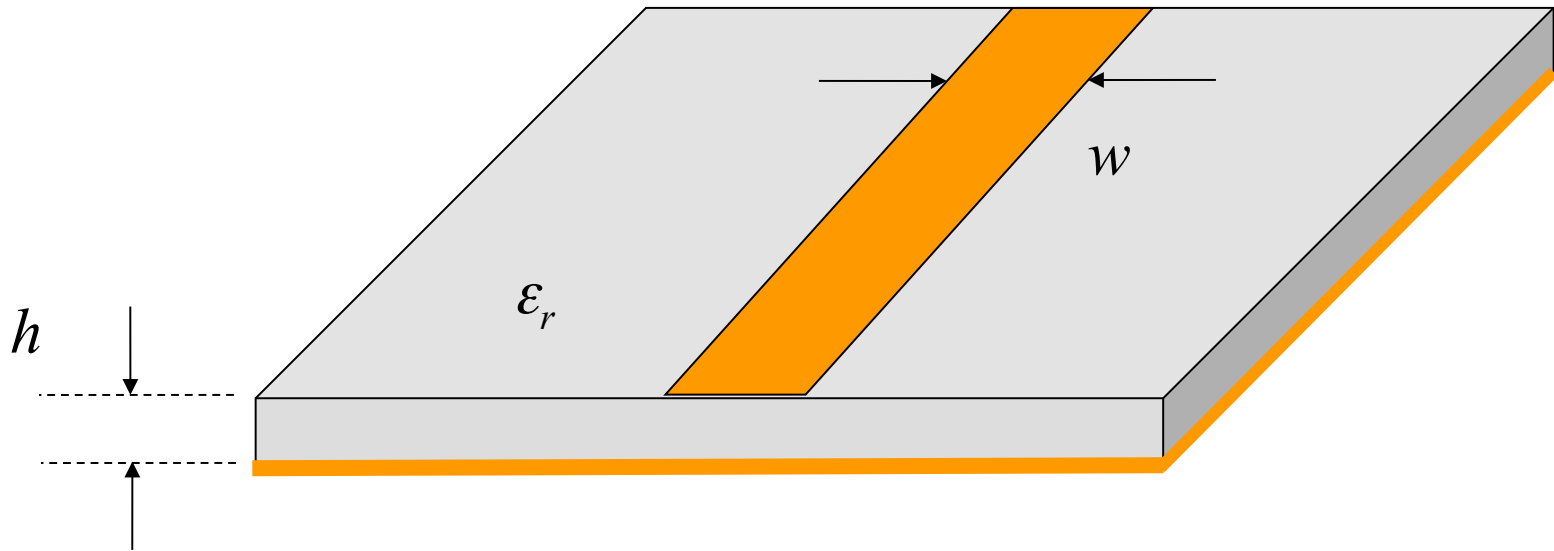
$$Z_0 = 75 \text{ } [\Omega]$$



Twin line

$$Z_0 = 300 \text{ } [\Omega]$$

Microstrip Line



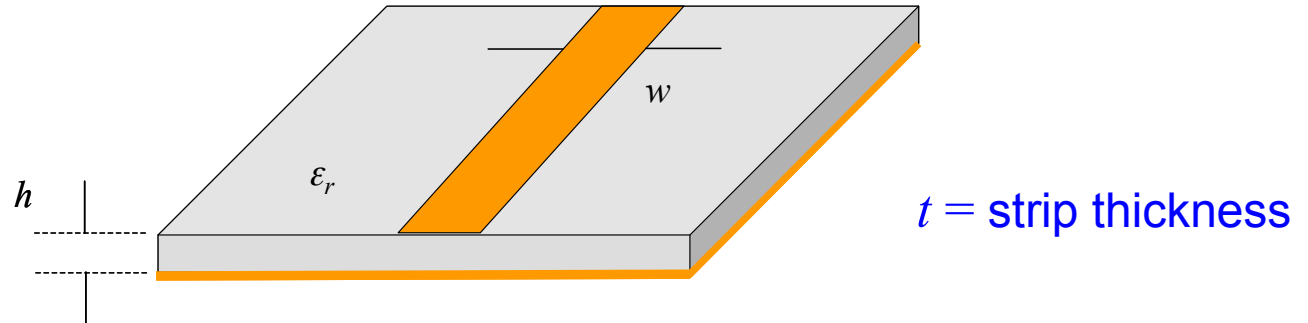
Parallel-plate formulas:

$$C \approx \epsilon_0 \epsilon_r \frac{w}{h}, \quad w \gg h$$

$$L \approx \mu_0 \frac{h}{w}, \quad w \gg h$$

$$Z_0 \approx \eta_0 \frac{1}{\sqrt{\epsilon_r}} \frac{h}{w}, \quad w \gg h$$

Microstrip Line (cont.)



More accurate CAD formulas:

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_r^{\text{eff}}} \left[(w'/h) + 1.393 + 0.667 \ln((w'/h) + 1.444) \right]} \quad (w/h \geq 1)$$

$$\epsilon_r^{\text{eff}} = \frac{\epsilon_r + 1}{2} + \left(\frac{\epsilon_r - 1}{2} \right) \left(\frac{1}{\sqrt{1 + 12(h/w)}} \right) - \left(\frac{\epsilon_r - 1}{4.6} \right) \left(\frac{t/h}{\sqrt{w/h}} \right) \quad (w/h \geq 1)$$

$$w' = w + \frac{t}{\pi} \left(1 + \ln \left(\frac{2h}{t} \right) \right)$$